Yining (Carina) Fu

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Professional Summary

Data-driven and strategic problem-solver with a Master's in Biostatistics and a minor in Economics. Proven track record of leveraging advanced analytics (SQL, R, Python, SAS) to translate complex quantitative models into actionable insights for cross-functional teams and senior stakeholders. Adept at applying rigorous experimental design, data modeling, and spatial clustering to drive process efficiency and inform strategic decision-making in healthcare, environmental, and financial sectors.

Education

Emory University — Atlanta, GA
Master of Public Health in Biostatistics — *Aug. 2024 – May. 2026*

Relevant Coursework: Biostatistics, Regression Analysis, Epidemiologic Methods, Machine Learning, Clinical Trials, Statistical Computing, Categorical Data Analysis

North Carolina State University — Raleigh, NC
Bachelor of Science in Statistics; Minor in Economics — *Aug. 2019 – Dec. 2023*

Honors: Dean's Honors List, 4 semesters

Experience

Emory University — Atlanta, GA
Quantitative Research Assistant — *Sep. 2025 – May. 2026*

- Engineered a robust data pipeline integrating 20 multidimensional socio-economic and environmental indicators (e.g., poverty, employment rates, housing stability, PM2.5, and population density); implemented log-transformation and reverse-coding to standardize vulnerability metrics across diverse neighborhood archetypes.
- Optimized advanced machine learning models (Self-Organizing Maps) using rigorous validation metrics (Elbow Method, Gap Statistic, Silhouette Analysis) to quantify cluster separation and evaluate localized environmental risks.
- Interpreted multidimensional socioeconomic patterns and assessed model classification robustness, while identifying temporal data limitations; delivered a fully documented dataset and intuitive visual reports to inform targeted public health intervention strategies for cross-functional clinical teams.

Chinese Academy of Sciences, Institute of Soil Science — Nanjing, China
Research Assistant — *Jan. 2024 – Jun. 2024*

- Built spatial segmentation and clustering models in SAS to identify critical regional contamination patterns, ensuring 100% data integrity through rigorous normalization and quality control protocols.
- Conducted data normalization, quality control, and exploratory analysis to prepare environmental datasets for statistical modeling.
- Supported quantitative interpretation of soil contamination trends for regional environmental assessment.

Shanxi Securities Research Institute — Shanghai, China
Industry Intern — *Jul. 2022 – Aug. 2022*

- Conducted fundamental analysis on the financial performance and valuation of publicly listed companies using Wind and Choice databases to support strategic equity research and investment decisions.
- Engineered automated Python and R scripts to calculate key valuation metrics (P/E, P/B) for 500+ companies, reducing manual data processing time by 30% and significantly enhancing analytical efficiency for the team.
- Synthesized complex financial data into streamlined Excel dashboards and PowerPoint presentations, tailoring data-driven recommendations for diverse audiences ranging from peer analysts to senior stakeholders.

Academic Projects

Aedes-Borne Viruses (ABV) Transmission Analysis Project — R

- Analyzed data from a large-scale cluster-randomized trial in Mexico using a Test-Negative Design; applied rigorous A/B testing principles to evaluate the efficacy of targeted interventions (Targeted Indoor Residual Spraying) against Dengue transmission.

- Processed and segmented the 2023 transmission dataset into distinct temporal phases (July–December) to conduct time-series and sensitivity analyses, effectively isolating true intervention impacts from general seasonal febrile trends.
- Developed custom R algorithms to calculate spatial clustering statistics, quantifying localized ”neighborhood risk” to establish intervention effectiveness; provided data-driven evidence that can be leveraged to optimize resource allocation for future outbreak prevention.

FAERS Machine Learning Project R

- Cleaned and merged seven FDA Adverse Event Reporting System datasets to build an analysis-ready dataset for machine learning applications.
- Developed and compared LDA, QDA, and KNN classification models to predict adverse event outcomes, achieving 80% prediction accuracy.
- Evaluated model performance and conducted gap analysis to identify limitations and potential areas for improvement.

Clinical Trials Case Study SAS

- Used SAS PROC SQL to clean and manipulate clinical trial datasets for efficacy analysis.
- Analyzed patient-level characteristics associated with clinical outcomes and treatment effectiveness.

Prospective Evaluation of Radial Keratotomy (PERK) Group Project R / SAS

- Conducted t-tests and ANOVA to assess relationships between patient characteristics and surgical outcomes.
- Produced numerical and graphical summaries to report post-surgical outcome improvements and procedure effectiveness.

Skills

Programming & Data Tools: R, Python, SAS, SQL, Java

Statistical Methods: Regression Analysis, Hypothesis Testing, ANOVA, t-tests, Missing Data Imputation, Classification, Clustering, A/B Testing

Packages & Platforms: Pandas, NumPy, ggplot2, dplyr, SAS Macro, SAS PROC SQL, Tableau, Microsoft Excel, PowerPoint

Languages: Mandarin Chinese, English